Convergence Analysis of Stochastic Gradient Descent with Adaptive Learning Rates for Non-Convex Objectives

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Abstract

We analyze the convergence properties of stochastic gradient descent (SGD) with adaptive learning rate schedules for optimizing non-convex loss functions encountered in deep learning. Let $f : \mathbb{R}^d \to \mathbb{R}$ be an L-smooth, non-convex function. We prove that with a learning rate schedule $\eta_t = \eta_0/\sqrt{t}$, vanilla SGD satisfies

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\big[\|\nabla f(\boldsymbol{\theta}_t)\|^2\big] \;\le\; \mathcal{O}\!\left(\frac{1}{\sqrt{T}}\right),$$

and that the Adam optimizer under mild assumptions achieves a convergence rate of $\mathcal{O}(1/\sqrt{T})$ in the same class of functions. We then derive a tighter bound under gradient sparsity assumptions common in transformer architectures and validate theoretical predictions empirically on a ResNet-18 trained on CIFAR-10.

1 Introduction and Problem Setup

Let $\mathcal{D}$ be a data distribution and let

$$f(\boldsymbol{\theta}) \;=\; \mathbb{E}_{(\boldsymbol{x},y)\sim\mathcal{D}}\big[\,\ell(h_{\boldsymbol{\theta}}(\boldsymbol{x}),\, y)\big]$$

denote the population risk, where $h_{\boldsymbol{\theta}} : \mathbb{R}^n \to \mathbb{R}^k$ is a neural network parameterized by $\boldsymbol{\theta} \in \mathbb{R}^d$ and $\ell : \mathbb{R}^k \times \mathcal{Y} \to \mathbb{R}_{\ge 0}$ is a loss function.

In practice we minimize the empirical risk

$$\hat{f}(\boldsymbol{\theta}) \;=\; \frac{1}{n}\sum_{i=1}^{n}\ell(h_{\boldsymbol{\theta}}(\boldsymbol{x}_i),\, y_i)$$

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using SGD updates of the form

$$\boldsymbol{\theta}_{t+1} \;=\; \boldsymbol{\theta}_t \;-\; \eta_t \widehat{\nabla f}(\boldsymbol{\theta}_t), \tag{1}$$

where $\widehat{\nabla f}(\boldsymbol{\theta}_t)$ denotes a mini-batch gradient estimate computed on a batch $\mathcal{B}_t \subset [n]$ of size b.

Definition 1 (L-Smoothness)**.** *A differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ is L-smooth if its gradient is L-Lipschitz continuous, i.e., there exists $L > 0$ such that for all $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^d$,*

$$\|\nabla f(\boldsymbol{u}) - \nabla f(\boldsymbol{v})\| \;\le\; L\,\|\boldsymbol{u} - \boldsymbol{v}\|.$$

An immediate consequence of L-smoothness is the descent lemma:

Lemma 1 (Descent Lemma)**.** *If f is L-smooth, then for any $\boldsymbol{\theta}, \boldsymbol{\phi} \in \mathbb{R}^d$,*

$$f(\boldsymbol{\phi}) \;\le\; f(\boldsymbol{\theta}) + \nabla f(\boldsymbol{\theta})^\top(\boldsymbol{\phi} - \boldsymbol{\theta}) + \frac{L}{2}\|\boldsymbol{\phi} - \boldsymbol{\theta}\|^2.$$

2 Convergence of SGD for Non-Convex Functions

Throughout this section we make the following standard assumptions.

Assumption A1 (Bounded variance): The stochastic gradient satisfies

$$\mathbb{E}\Big[\widehat{\nabla f}(\boldsymbol{\theta})\Big] = \nabla f(\boldsymbol{\theta}), \qquad \mathbb{E}\Big[\|\widehat{\nabla f}(\boldsymbol{\theta}) - \nabla f(\boldsymbol{\theta})\|^2\Big] \le \sigma^2.$$

Assumption A2 (Lower bounded): $f^* := \inf_{\boldsymbol{\theta}} f(\boldsymbol{\theta}) > -\infty$.

Theorem 1 (SGD Convergence, Non-Convex)**.** *Under Assumptions A1–A2 and L-smoothness of f, with learning rate $\eta_t = \eta/\sqrt{T}$ for a fixed $\eta > 0$, the iterates* (1) *satisfy*

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\big[\|\nabla f(\boldsymbol{\theta}_t)\|^2\big] \;\le\; \frac{f(\boldsymbol{\theta}_1) - f^*}{\eta\sqrt{T}} + \frac{\eta L \sigma^2}{2\sqrt{T}} \;=\; \mathcal{O}\!\left(\frac{1}{\sqrt{T}}\right).$$

Proof. Applying the descent lemma with $\boldsymbol{\phi} = \boldsymbol{\theta}_{t+1}$:

$$f(\boldsymbol{\theta}_{t+1}) \le f(\boldsymbol{\theta}_t) - \eta_t \nabla f(\boldsymbol{\theta}_t)^\top \widehat{\nabla f}(\boldsymbol{\theta}_t) + \frac{L\eta_t^2}{2}\|\widehat{\nabla f}(\boldsymbol{\theta}_t)\|^2.$$

Taking expectations and using A1:

$$\mathbb{E}[f(\boldsymbol{\theta}_{t+1})] \le \mathbb{E}[f(\boldsymbol{\theta}_t)] - \eta_t\,\mathbb{E}\big[\|\nabla f(\boldsymbol{\theta}_t)\|^2\big] + \frac{L\eta_t^2}{2}\big(\mathbb{E}\big[\|\nabla f(\boldsymbol{\theta}_t)\|^2\big] + \sigma^2\big).$$

Rearranging and summing from $t = 1$ to T with $\eta_t = \eta/\sqrt{T}$ yields the claim. □ □

3 The Adam Optimizer

Adam (Kingma and Ba, 2015) maintains running estimates of the first and second moments:

$$\boldsymbol{m}_t = \beta_1 \boldsymbol{m}_{t-1} + (1-\beta_1)\,\widehat{\nabla f}(\boldsymbol{\theta}_t) \tag{2}$$

$$\boldsymbol{v}_t = \beta_2 \boldsymbol{v}_{t-1} + (1-\beta_2)\,\widehat{\nabla f}(\boldsymbol{\theta}_t)^{\odot 2} \tag{3}$$

with bias-corrected estimates $\hat{\boldsymbol{m}}_t = \boldsymbol{m}_t/(1-\beta_1^t)$ and $\hat{\boldsymbol{v}}_t = \boldsymbol{v}_t/(1-\beta_2^t)$. The parameter update is

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \frac{\eta}{\sqrt{\hat{\boldsymbol{v}}_t} + \epsilon} \odot \hat{\boldsymbol{m}}_t, \tag{4}$$

where $\odot$ denotes element-wise multiplication and division.

Proposition 1 (Adam Convergence, Simplified)**.** *Under A1–A2, L-smoothness, and bounded gradients $\|\nabla f(\boldsymbol{\theta})\| \le G$, Adam with $\eta_t = \eta/\sqrt{t}$ satisfies*

$$\min_{1\le t\le T} \mathbb{E}\big[\|\nabla f(\boldsymbol{\theta}_t)\|^2\big] \;\le\; \mathcal{O}\!\left(\frac{G\, d^{1/2} \ln T}{\sqrt{T}}\right).$$

The additional $\ln T$ factor relative to vanilla SGD arises from the adaptive preconditioner $\hat{\boldsymbol{v}}_t^{-1/2}$ and is unavoidable in the worst case.

4 Tighter Bounds Under Gradient Sparsity

In transformer architectures, gradients are approximately s-sparse: at most $s \ll d$ coordinates have large magnitude at each step. Formally, we assume there exists an index set $S_t \subseteq [d]$ with $|S_t| = s$ such that

$$\|\nabla f(\boldsymbol{\theta}_t) - [\nabla f(\boldsymbol{\theta}_t)]_{S_t}\| \;\le\; \delta\, \|\nabla f(\boldsymbol{\theta}_t)\|$$

for some small $\delta \in (0,1)$, where $[\cdot]_{S_t}$ retains only coordinates in S_t.

Under this sparsity assumption, the effective dimension is s rather than d, giving a tighter convergence bound for Adam:

$$\min_{1\le t\le T} \mathbb{E}\big[\|\nabla f(\boldsymbol{\theta}_t)\|^2\big] \;\le\; \mathcal{O}\!\left(\frac{G\, s^{1/2} \ln T}{\sqrt{T}}\right).$$

For typical transformer gradients where $s/d \approx 0.01$, this represents a $\sqrt{d/s} \approx 10\times$ improvement over the non-sparse bound.

5 Empirical Validation

Table 1 reports final test accuracy and convergence round (number of gradient steps to reach 90% of final accuracy) on CIFAR-10 with ResNet-18 under four optimization configurations.

The empirical convergence speeds match the theoretical ordering: Adam with warmup converges fastest, consistent with the tighter $\mathcal{O}(1/\sqrt{T})$ bound under the adaptive preconditioner combined with a well-tuned schedule.

Table 1: Optimization results on CIFAR-10 (ResNet-18, mean $\pm$ std over 5 seeds)

Optimizer	η_0	**Schedule**	**Test Acc. (%)**	**Steps to 90%**
SGD (fixed η)	0.1	constant	91.2 ± 0.3	45,200
SGD + cosine decay	0.1	cosine	93.4 ± 0.2	38,700
Adam	3×10^{-4}	constant	92.1 ± 0.4	31,900
Adam + warmup	3×10^{-4}	linear warmup + decay	$\mathbf{94.1 \pm 0.2}$	**28,400**

6 Conclusion

We have established $\mathcal{O}(1/\sqrt{T})$ convergence for SGD and Adam on L-smooth non-convex objectives and derived a $\sqrt{d/s}$-tighter bound for Adam under gradient sparsity. These results provide theoretical grounding for the practical preference for adaptive optimizers in deep learning and motivate future work on exploiting sparsity structure for communication-efficient distributed training.

References

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